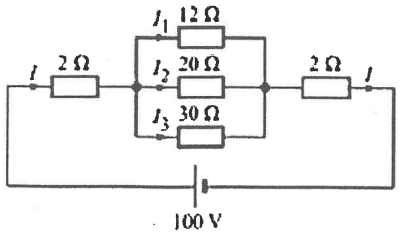
USN | | | | | | | | | | [23CSE108]

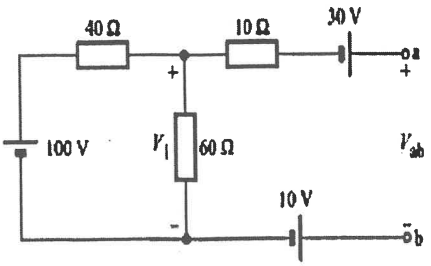
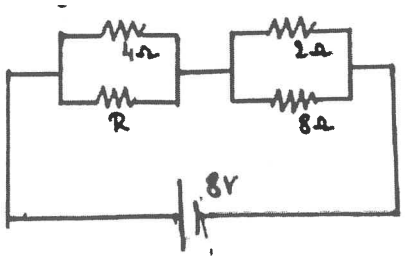
FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024
FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Draw the necessary circuit diagram as applicable.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Compare Resistance and Conductance.	L-2	CO 1	**02**
	b.	What is Form Factor and Peak Factor?	L-1	CO 2	**02**
	c.	What is Voltage Regulation in a Transformer?	L-1	CO 3	**02**
	d.	Draw the V-I characteristic of a P-N Junction Diode.	L-2	CO 4	**02**
	e.	Explain the term rectification and state its application.	L-2	CO 5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Determine the currents I_1, I_2, I_3 flowing in the parallel branches in the network shown below 	L-4	CO 1	**05**
		OR			
	b.	Define series resistor circuit with circuit diagram and explain voltage division in series resistor circuit.	L-2	CO 1	**05**
3.	**a.**	Explain the significance of the term Resonance in a series **R-L-C** circuit. Give mathematical formula and phasor diagram for Resonance.	L-3	CO 2	**05**
		OR			
	b.	Derive the expression for voltage, current, and power for a single-phase AC circuit containing **R-L** connected in series. Draw the necessary waveform and phasor diagram.	L-3	CO 2	**05**
4.	**a.**	What is an Ideal Transformer? Explain its characteristics with the required diagram.		CO 3	**05**
		OR			
	b.	The primary of a 50 Hz, step-down transformer has 480 turns and it is fed from a 6400V supply. Find **i.** The peak value of the flux produced. **ii.** The voltage across the secondary winding if it has 20	L-4	CO 3	**05**

		turns.			
5.	**a.**	Draw and explain the forward bias characteristics of the PN junction diode.	L-2	CO 4	**05**
		OR			
	b.	Explain Avalanche breakdown with the necessary graph.	L-2	CO 4	**05**
6.	**a.**	Describe the operation of a centre-tapped full wave rectifier.	L-2	CO 5	**05**
		OR			
	b.	Explain the working of a half-wave rectifier with a neat circuit diagram and waveforms.	L-2	CO 5	**05**
7.	**a.**	Define the efficiency of the transformer. Derive the condition for maximum efficiency of the transformer.	L-3	CO 3	**05**
		OR			
	b.	Define Real, Reactive and Apparent Power with the help of power triangle in AC Circuits. Also, derive power factor with the help of power triangle.	L-2	CO 2	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Find the voltage $\mathbf{V_{ab}}$ across the open circuit in the circuit shown in figure below: 	L-5	CO 1	**10**
		OR			
	b.	If the total power dissipated in the network shown below is 16 Watt. Find the value of 'R' and the total current. 	L-5	CO 1	**10**
9.	**a.**	Explain in detail about the relation between voltage, current and power in RLC series circuit, for different cases. Draw the phasor diagrams.	L-3	CO 2	**10**

		OR			
	b.	A 230 V , 50 Hz ac supply is applied to a coil of 0.06 H inductance and 2.5 Ω resistance connected in series with a 6.8 μF capacitor. Calculate **i.** Impedance **ii.** Current **iii.** Phase angle between current and voltage **iv.** Power factor **v.** Power consumed	L-5	CO 2	**10**
10.	**a.**	Discuss in detail about the major components of a transformer and briefly explain the losses occurring in each component of transformer.	L-3	CO 3	**10**
		OR			
	b.	What is transformation ratio of a transformer? Classify transformer based on construction.	L-2	CO 3	**10**
11.	**a.**	Explain the working of a Bridge rectifier with relevant waveforms and calculate its efficiency and ripple factor.	L-3	CO 5	**10**
		OR			
	b.	Differentiate between PN Junction and Zener Diode. Explain the working of a Zener Diode as voltage regulator.	L-2	CO 5	**10**
12.	**a.**	Explain the input and output characteristics of a transistor under common emitter mode configuration.	L-2	CO 4	**10**
		OR			
	b.	Explain the principle of an NPN transistor and its operation modes.	L-2	CO 4	**10**
13.	**a.**	State and explain Kirchoff's Laws, and list its advantages, disadvantages, and applications.	L-1	CO 1	**10**
		OR			
	b.	Define Average value and RMS value. Derive an expression for Average and RMS value of a sinusoidal current.	L-4	CO 2	**10**